

# WiViHAR: A Deep Learning-Based Human Activity Recognition Method Using WiFi and Vision Multimodal Fusion

Xinyu Ma , Yankan Yang , and Kun Bian

**Abstract**—Vision-based human activity recognition (HAR) techniques have garnered significant attention and extensive research, with successful applications in many scenarios. However, their recognition accuracy deteriorates significantly under conditions, such as lighting changes and object occlusion. Considering that WiFi-based HAR techniques are unaffected by lighting conditions, we propose WiViHAR, a multimodal HAR method by fusing WiFi and vision information. Specifically, this method first utilizes atrous spatial pyramid pooling (ASPP) module and convolutional block attention module to design a dedicated deep learning network for processing WiFi channel state information. Concurrently, frame images are extracted and resized from vision, and the ASPP is extended to 3-D structure to design a dedicated deep learning network for processing the vision information. Finally, we concatenate the outputs of the two networks and apply the random forest decision fusion method to leverage the advantages of both WiFi and vision, enhancing the environmental adaptability of the method. Extensive experiments conducted in real-world environments demonstrate that the proposed method significantly outperforms single-modal WiFi or vision-based HAR methods, achieving an accuracy of over 98.57% even under lighting changes or occlusions. This highlights the robustness of the proposed method, offering a novel approach for HAR fields.

**Index Terms**—Channel state information (CSI), human activity recognition (HAR), multimodal fusion, vision, WiFi.

## I. INTRODUCTION

**H**UMAN activity recognition (HAR) technology has been widely applied across various domains, including smart homes, healthcare, security monitoring, autonomous driving, and human-computer interaction [1]. For example, in elderly care, HAR enables real-time monitoring and timely alerts for anomalies, such as falls, reducing associated risks [2].

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In intelligent systems, recognizing human actions improves intent interpretation and interaction efficiency, enhancing artificial intelligence practicality [3].

HAR methods are mainly divided into vision-based and wireless signal-based methods. Traditional vision-based methods rely on hand-designed feature extraction and classifiers, while deep learning models in recent years have significantly improved recognition accuracy of the methods by enabling automatic feature extraction. However, challenges remain in privacy protection and environmental adaptability, especially under low-light or occlusion conditions [4]. To improve vision coverage, some researchers have used multiple cameras, but this raises privacy concerns and fails under low-light conditions. Others have explored depth cameras [5] or thermal infrared cameras [6], which enhance performance but are costly and ineffective against occlusion conditions. Deep learning-based enhancement techniques offer partial improvements, but remain limited with occlusion conditions.

Wireless signal-based HAR, particularly WiFi-based methods, have attracted increasing attention due to their independence from lighting conditions and enhanced privacy. Compared with the vision-based methods, WiFi offers advantages such as non-line-of-sight detection, device-free, low cost, and easy deployment [7]. However, WiFi signals are susceptible to interference from environmental changes and human activities, making it challenging to achieve high accuracy. Specifically, we referenced the experimental method of Zhang et al. [8] and used the maximum mean discrepancy (MMD) to quantify WiFi signals' distribution differences in different environmental layouts. When the layout remains unchanged, the MMD value is below 0.02, indicating consistent signal distribution. When the layout changes slightly, the MMD value exceeds 0.2, indicating a certain difference. When the layout changes significantly, the MMD value approaches 0.5, indicating a large difference. This confirms that WiFi signals are highly sensitive to environmental changes, and it can also be inferred that they are susceptible to interference from human activities. In contrast, the vision-based HAR methods have shown strong performance in these areas and can provide substantial help.

Therefore, HAR technology based on the fusion of WiFi and vision has garnered significant attention from researchers. However, there are currently few related studies, and they often focus on feature fusion [9], [10]. Fusion methods are typically categorized into feature fusion, decision fusion, and hybrid

fusion. Feature fusion refers to integrate features from all modalities after feature extraction but before modality modeling. Although this method is relatively simple and can exploit cross-modal correlations, it is highly sensitive to data noise and modality degradation. In contrast, decision fusion allows independently modeling each modality and combining the outputs or decisions of the individual models to generate the final result, offering greater robustness against data noise and modality degradation. Hybrid fusion combines the advantages of feature fusion and decision fusion, but it also suffers from issues, such as complex structure and high computational cost.

To address these challenges, we propose WiViHAR, a WiFi and vision multimodal fusion HAR method based on improved deep learning. This method uses commercial off-the-shelf (COTS) WiFi devices and high-definition cameras, including a WiFi sensing module, a vision sensing module, and a multimodal fusion module. In the WiFi sensing module, we designed a specialized WiFi deep learning network, including convolutional layers, two slightly different atrous spatial pyramid pooling (ASPP) modules, and a convolutional block attention module (CBAM). This specialized design enhances the perception of multiscale information and improves the focus and discrimination of important features. In the vision sensing module, we designed a specialized vision deep learning network, including 3-D convolutional layers and two ASPP modules extended to 3-D structures, enabling the extraction of more comprehensive spatiotemporal information. In the multimodal fusion module, we designed a novel decision fusion method to concatenate the outputs of the WiFi and vision modules, and input into a random forest (RF) method to obtain the final recognition result. Compared with other fusion methods, decision fusion is more robust against data noise and modality degradation, while achieving a good balance between recognition performance and computational cost, making it our first choice. Extensive experiments conducted in a real classroom environment demonstrate that even under conditions of lighting changes or occlusion, the accuracy of our proposed method exceeds 98.86%, significantly outperforming single-modality methods using either WiFi or vision alone and demonstrating its strong robustness.

The contributions of this article are summarized as follows.

- 1) We propose WiViHAR, a multimodal fusion method for HAR based on WiFi and vision. The complementary advantages of the WiFi and vision are leveraged through decision fusion so as to achieve superior performance. This approach offers a novel perspective for the development of the multimodal fusion field and the activity recognition field.
- 2) We design specialized deep learning network models for WiFi and vision, verifying the effectiveness of the ASPP module in activity recognition and extending it to a 3-D structure. Ablation experiments demonstrate the effectiveness of these specialized network models.
- 3) We conducted extensive experiments in a real classroom. Despite various environmental changes, our proposed method still significantly outperforms single-modality methods using either WiFi or vision alone.

The rest of this article is organized as follows. Section II reviews related work. Section III describes our proposed method in detail. Section IV presents the experimental settings, results, and discussions. Finally, Section V concludes this article.

## II. RELATED WORK

### A. Vision

As early as the 1970s, Marr [11] proposed the theory of computer vision, initiating the research on activity recognition. According to the different ways of extracting features, these studies can be categorized as: traditional handcrafted feature extraction methods and deep learning-based methods. However, the former relies on relatively small datasets and lack generality, making it insufficient for modern HAR tasks.

Simonyan and Zisserman [12] proposed a two-stream convolutional network, extracting spatiotemporal features from RGB images and optical flow information. Xing et al. [13] introduced a two-stream multiscale HAR and anticipation network based on temporal channel attention, enhancing the model's ability to express temporal and spatial features. Tran et al. [14] developed the C3D network for HAR, using 3-D convolutions to capture different dimensional features of both time and space. Dong et al. [15] proposed an attention residual 3-D network, improved the ordinary 3-D residual structure, and designed a 3-D spatiotemporal attention mechanism module. Dosovitskiy et al. [16] introduced a Vision Transformer architecture to eliminate the traditional reliance on convolutional neural networks (CNNs). Lin et al. [17] developed a cross attention and built a cross-attention transformer network to achieve better performance in image recognition. Although the deep learning-based methods have achieved great success, their recognition accuracy will be significantly affected under low-light or object occlusion conditions.

### B. WiFi

WiFi-based HAR leverages the varying degrees of interference of WiFi signals caused by human movements to identify the associated human activities. There are two main methods: using WiFi received signal strength indication (RSSI) and using WiFi channel state information (CSI). However, the RSSI is coarse-grained and unstable, which has great limitations in accurately identifying human activity.

Wang et al. [18] proposed the E-eyes system, using Earth mover's distance and dynamic time warping to match CSI amplitude templates respectively. Wang et al. [19] introduced the CSI based human activity recognition and monitoring system (CARM), established a CSI speed model and a CSI activity model, and quantified the correlation between CSI value dynamics and a specific human activity. Zhang et al. [20] proposed a Fresnel zones-based repetitive activity recognition method, which recognizes and counts through periodic changes in CSI amplitude. Lin et al. [21] developed an Android smartphone application to collect and extract CSI through firmware patches, further reducing costs and enhancing convenience. Although WiFi CSI can achieve finer-grained activity recognition, it is

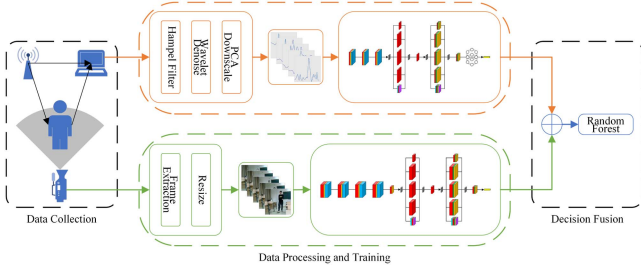


Fig. 1. Overall framework of WiViHAR.

easily affected by environmental changes and human activities. Therefore, combining vision information and WiFi CSI for multimodal fusion recognition can compensate for their respective shortcomings, thereby enhancing the robustness of HAR.

### C. Multimodal Fusion

Multimodal fusion refers to the process of integrating information from two or more modalities to make predictions. It can supplement missing information and enhance the robustness of prediction models [22]. This article specifically focuses on WiFi and vision-based multimodal fusion research.

Deng et al. [9] proposed a multimodal gait recognition method, GaitFi, using a lightweight residual convolutional network as the backbone network to integrate WiFi and vision features. Hou et al. [23] introduced a multimodal emotion recognition method and designed a multisource learning framework to combine WiFi and vision features. Li et al. [10] proposed a multimodal activity recognition method to design a long short-term transformer network to establish cooperation between CSI data and skeleton points. Yang et al. [24] developed an unsupervised multimodal activity recognition method, MaskFi, devising a masking WiFi-vision modeling method to learn cross-modal and single-modal features. In summary, most current fusion methods focus on feature fusion. This is difficult to adapt to drastic changes in a single modality, resulting in poor stability.

## III. METHOD

To address the limitations of the current multimodal fusion methods, we propose a novel multimodal fusion method utilizing the deep learning network. The overall framework of our method is illustrated in Fig. 1 and consists of three main components: data collection, data processing and training, and decision fusion. First, data are synchronously collected using WiFi devices and cameras. Second, the data are preprocessed separately, followed by training with the designed network models. Finally, the outputs of the two models are fused using an RF method to obtain the final result.

### A. WiFi Sensing Module

1) *Theoretical Foundation*: In typical indoor environments, wireless signals propagate through multiple paths, such as direct, reflection, refraction, and scattering, and carry information reflecting environmental characteristics in a multipath superposition manner [25]. When the signal encounters the human

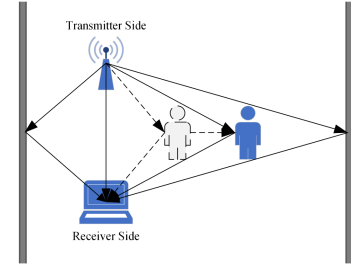


Fig. 2. WiFi signal propagation model.

body during propagation, it will cause changes to the normal signal, thereby can recognized human activities. The WiFi signal propagation model is illustrated in Fig. 2.

CSI provides a highly fine-grained description of the state attributes of wireless communication link channels and accurately reflects the information within the indoor spatial environment. CSI measures not only the frequency response of multiple subcarriers but also the amplitude and phase information of each subcarrier. They can be obtained simply by using a wireless network card and open-source firmware.

In the frequency domain, the model of CSI is as follows:

$$Y = HX + N \quad (1)$$

where  $H$  represents the CSI channel matrix,  $Y$  is the received signal vector,  $X$  is the transmitted signal vector, and  $N$  is the noise vector. Based on the equation above, the channel matrix  $H$  can be estimated as

$$H = Y/X. \quad (2)$$

Using the Intel 5300 wireless card, packets containing 30 subcarriers can be obtained, with a set of each packet's CSI represented as

$$H = [H_1, H_2, \dots, H_k, \dots, H_{30}] \quad (3)$$

where  $H_k$  represents the CSI of the  $k$ th subcarrier. The CSI of a single subcarrier can be represented as

$$H(k) = |H(k)|e^{j\sin\angle H(k)} \quad (4)$$

where  $|H(k)|$  represents the amplitude of the  $k$ th subcarrier, and  $\angle H(k)$  represents the phase of the  $k$ th subcarrier.

2) *WiFi Data Preprocessing*: Due to factors, such as environmental noise, electromagnetic interference, or changes in the internal state of devices, the received CSI signals may contain outliers. The Hampel filter [26] is used for outlier detection and replacement. It can filter out all outliers outside the interval  $[\mu - \lambda \times \sigma, \mu + \lambda \times \sigma]$ , where  $\mu$ ,  $\lambda$ , and  $\sigma$  represent the median, anomaly discrimination parameter, and median absolute deviation, respectively. Therefore, we first use the Hampel filter for the preprocessing of CSI data. Fig. 3(a) shows the original CSI signal waveform of a drinking water activity, and Fig. 3(b) shows the signal waveform after Hampel filtering. It is evident that the outliers are well filtered, but many minor fluctuations still remain, requiring further processing.

In order to further eliminate fluctuations from the CSI signal, we applied wavelet denoise [27] to it, which can exhibit good

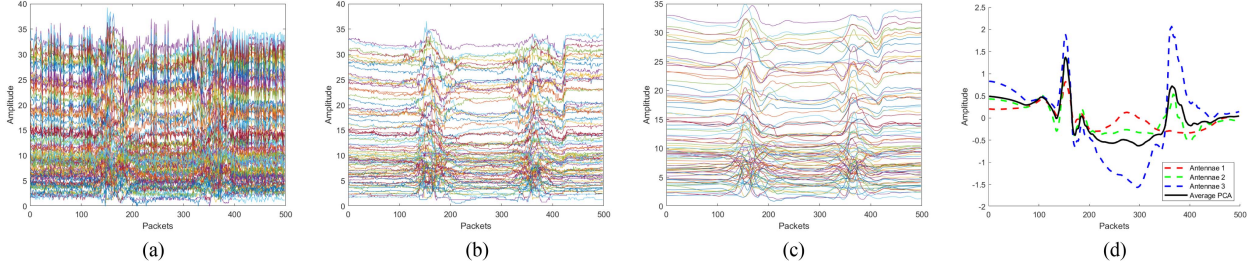


Fig. 3. WiFi data preprocessing. (a) Original signal waveform. (b) Signal waveform after Hampel filtering shows that although the mutation noise is well filtered, many minor fluctuations still remain and require further processing. (c) Signal waveform after wavelet denoising is smoother than that after Hampel filtering, while well preserving the original signal features. (d) PCA is performed separately on the signals from the three antennas, and the results are then averaged to retain the majority of the information.

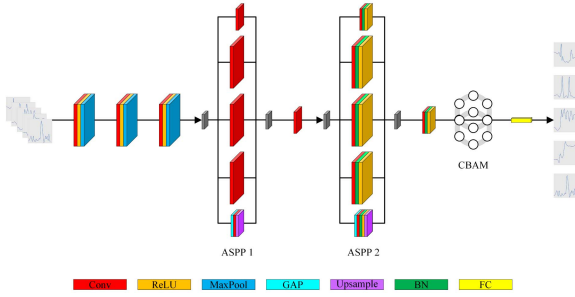


Fig. 4. WiFi deep learning model.

properties in multiresolution and time–frequency localization. The formula of the wavelet denoise is as follows:

$$W_{WT}(a, \tau) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} f(t) \mu\left(\frac{t-\tau}{a}\right) dt \quad (5)$$

where  $a$  is the scale of the wavelet function, and  $\tau$  is the translation parameter. Fig. 3(c) shows the WiFi signal waveform after wavelet denoising. It can be observed that the waveform is smoother than that after Hampel filtering, with the original signal features well preserved.

Since the received CSI signals contain multiple subcarriers from multiple antennas, it will lead to excessive data dimensions and complex features. Principal component analysis (PCA) [28] can project high-dimensional data into low-dimensional space through linear transformation and maximize the variance information of the data, thereby extracting the most representative signal features. Moreover, because each antennae's position varies slightly, it carries different feature information. Therefore, we perform PCA separately on the signals from the three antennas to select the first principal component from each, and then average them to obtain the majority of the information, as shown in Fig. 3(d).

3) *WiFi Deep Learning Model*: To utilize CSI information for HAR, we designed a WiFi deep learning model. This model comprises three convolutional layers, two partially similar ASPP modules and a CBAM, as depicted in Fig. 4.

In recent years, CNN have demonstrated powerful capabilities in extracting image features. However, traditional fixed receptive-field convolutional kernels have certain limitations. First, it is difficult to capture multiscale information, such as local and global contexts simultaneously; second, spatial details

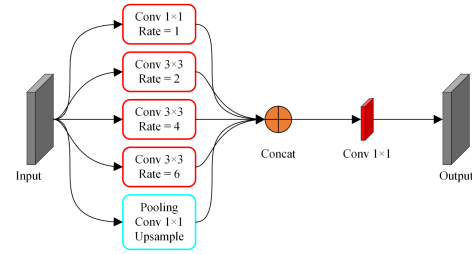


Fig. 5. Structure of ASPP.

may be lost during the downsampling process; third, if the receptive field is forcibly expanded, computing resources will increase significantly.

In order to effectively expand the receptive field of the convolutional kernel and enhance the model's ability to perceive multiscale information, we constructed an ASPP module. It was originally proposed in DeepLab v2 [29], which combines atrous convolution and multiscale feature fusion strategies, and can improve the model's performance and accuracy. Among them, atrous convolution expands the receptive field range by adjusting the dilation rate of convolutional kernels to capture features at different scales. In addition, ASPP introduces a global average pooling branch to capture global contextual information. The structure of ASPP is illustrated in Fig. 5, and its computational process is outlined as follows:

$$Y = \text{Concat}(C_{1,1}(X), C_{2,3}(X), C_{4,3}(X), C_{6,3}(X), P(X)). \quad (6)$$

In the equation,  $C_{m,n}(X)$  represents atrous convolution with a dilation rate of  $m$  and a kernel size of  $n \times n$  for  $X$ , while  $P(X)$  denotes the feature extraction using global average pooling followed by convolution and sampling operations. Concat represents the concatenation of all preceding components, preparing the features for output.

We utilized two ASPP modules, with the first module being identical to the structure described above, while the second module included batch normalization layers and activation functions after each convolutional layer. This addition enhances the stability of the model, increasing the model's expression capacity, and facilitating the learning of subsequent attention mechanisms.

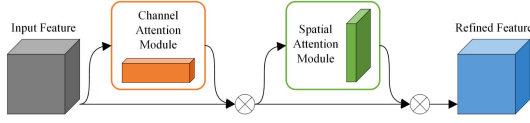


Fig. 6. Structure of CBAM.

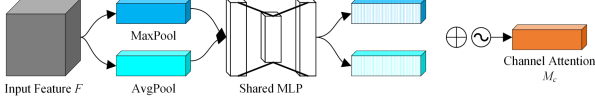


Fig. 7. Structure of CAM.

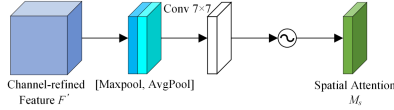


Fig. 8. Structure of SAM.

With the emergence of other excellent network models, certain structural flaws of CNN gradually become evident. First, it mainly relies on local receptive fields for feature extraction and lacks global information. Second, it cannot effectively distinguish between important and irrelevant information, and is easily disturbed. Third, it lacks a feature selection mechanism and cannot adaptively adjust feature weights.

In order to automatically learn the correlations between different spatial positions and channel dimensions in images, and to selectively extract the most important features in images, we constructed CBAM [30]. It consists of a channel attention module (CAM) and a spatial attention module (SAM), and can improve the model's feature learning and expression capabilities. The structure of CBAM is depicted in Fig. 6.

The CAM can adaptively adjust the weights of different channel features. It performs global max pooling and average pooling on feature maps of different channels, inputs them into a multilayer perceptron (MLP) with weight sharing, and processes them with element-by-element summarization and Sigmoid activation function. The CAM is illustrated in Fig. 7, and its computation formula is as follows:

$$M_c(F) = \sigma(W_1(W_0(F_{\text{avg}}^c) + W_1(W_0(F_{\text{max}}^c))))). \quad (7)$$

In the above equation,  $M_c$  represents the 1-D channel attention map,  $\sigma$  denotes the Sigmoid activation function,  $W_0$  and  $W_1$  are the weights of the two MLP layers,  $F_{\text{avg}}^c$  and  $F_{\text{max}}^c$ , respectively, represent the features after average pooling and max pooling along the channels, and  $r$  represents the reduction ratio of intermediate channels.

The SAM can determine the importance of different regions by analyzing spatial structures and positional information. It performs global max-pooling and average-pooling operations on the input feature map, compressing and concatenating it along the channel dimension. Then, a single convolutional layer is applied to adjust spatial dimensions. The SAM is illustrated in Fig. 8, and its computation formula is as follows:

$$M_s(F) = \sigma(f^{7 \times 7}([F_{\text{avg}}^s; F_{\text{max}}^s])). \quad (8)$$

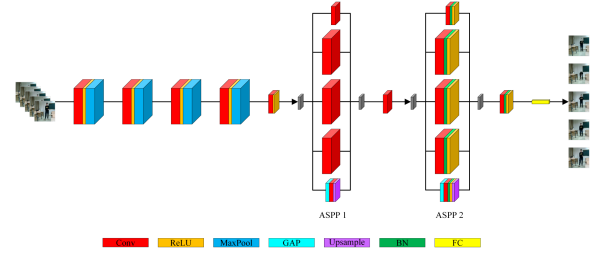


Fig. 9. Structure of vision deep learning model.

In the above equation,  $M_s$  represents the 2-D spatial attention map,  $\sigma$  denotes the Sigmoid activation function,  $f^{7 \times 7}$  refers to a convolutional kernel with a size of  $7 \times 7$ , and  $F_{\text{avg}}^s$  and  $F_{\text{max}}^s$  represent the features after average pooling and max pooling in the spatial dimension, respectively.

### B. Vision Sensing Module

In daily environments, raw vision information consists of a 5-s video captured at 30 frames per second, totaling 150 frames. In dark environments, the video changes to 15 frames per second, totaling 75 frames. It can be observed that the number of frames in the daily environment is twice that in the dark environment. Based on commonly used frame sampling settings and preliminary experimental validation, we choose to extract one frame every eight frames from the daily environment video, resulting in 19 frames, and one frame every four frames from the dark environment video, also resulting in 19 frames. This process standardizes the data length by maintaining consistent time intervals between frames under both conditions, thereby avoiding information loss and simplifying model design. This series of RGB frame images has an original size of  $1280 \times 720$ . The images are cropped to  $720 \times 720$  by removing excess parts on both sides, and then resized to  $180 \times 180$  as the final data.

The vision deep learning model consists primarily of five 3-D convolutional layers and two partially identical ASPP modules. Compared to the WiFi deep learning model, the two additional convolutional layers are added. This adjustment is mainly due to the fact that consecutive frame images are data with higher dimensional structures containing rich spatial and temporal information. More convolutional layers can provide deeper feature extraction capabilities, better representing information at different levels. The structure of the ASPP module is similar to that in the WiFi deep learning model, but extended to a 3-D structure. It can extract features in both temporal and spatial dimensions with multiple dilation rates to capture local details and spatiotemporal context information. This helps the model better understand the temporal relationships and spatial structures within consecutive frame images. The structure of the vision deep learning model is shown in Fig. 9.

### C. Multimodal Fusion Module

In the multimodal fusion module, assuming that there are  $k$  synchronized WiFi and vision output results in each batch and  $j$  action categories, each model will output a  $k \times j$  matrix. We concatenate the outputs of the two models, resulting in a

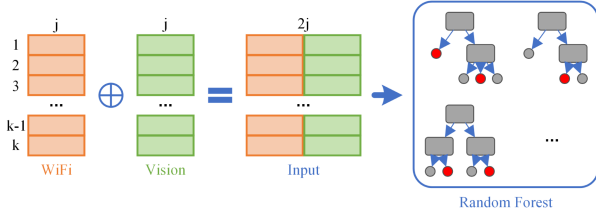


Fig. 10. Multimodal fusion.

$k \times 2j$  matrix as the input to the RF, as shown in Fig. 10. By using concatenation, on the one hand, we can comprehensively utilize the output results of both models, retaining all content and providing richer feature information. On the other hand, by concatenating the output results positionwise, the correlation between features is enhanced so as to capture the relationships between relevant features in the input data.

Common decision fusion methods can be categorized into mathematical computation methods, machine learning methods, and neural network methods. The mathematical computation methods include simple operations, such as addition, averaging, and weighting. Although these methods are easy to understand and implement, they heavily rely on the results of the preceding models. Upon analyzing the concatenated data, we find its overall simplicity. The neural network methods typically require sufficient samples and complexity to harness their powerful capabilities. When faced with simple data, it may be challenging for them to learn meaningful feature representations, and they are prone to overfitting. The machine learning methods often exhibit better adaptability to simple data. Among them, the RF method demonstrates strong robustness against noise and missing data, capable of handling incomplete or noisy data to a certain extent. Specifically, each decision tree in the RF is constructed based on randomly selected training samples and feature subsets, ensuring differences among them. Then, the results of individual decision trees are integrated through majority voting so as to enhance the model's diversity and robustness [31].

#### IV. EXPERIMENTATION

##### A. Experimental Setup

1) *Experimental Environment*: The experimental environment was set up in a typical university classroom with fixed tables and chairs. The router and the receiving antenna are placed on the opposite sides of the aisle, and the camera is located in the diagonal front, as shown in Fig. 11(a). The experimental transmitter is a TP-Link 886 N wireless router, and the receiver is a ThinkPad X200 laptop with an Intel 5300 wireless NIC and equipped with three omni-directional antennas. The data acquisition frequency is set to 100 Hz. The video data are collected by an Aoni A25 high-definition camera synchronizing with the CSI data collection.

According to the Fresnel zone theory [32], it can be seen that the energy of the CSI signal is mainly concentrated in the first 8–12 zones in the propagation process, and more than 70% of the energy is transmitted in the first zone. In order to achieve good experimental results, we designed most of the activities

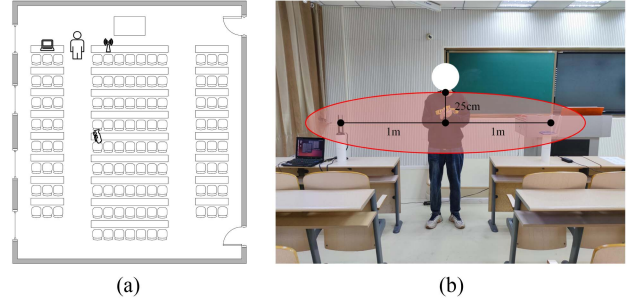


Fig. 11. Experimental environment. (a) Floor plan of classroom. (b) Experimental scene.

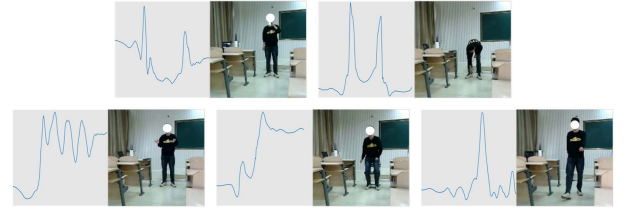


Fig. 12. Daily environment.



Fig. 13. Dark environment.

range within the first zone. According to the Fresnel zone radius formula, we set the distance between the transmitting antenna and the receiving antenna to be 2 m, and the router works in the 2.4-GHz band, and we can calculate the radius of the first Fresnel zone to be 25 cm, as shown in Fig. 11(b).

2) *Dataset*: The experiment data were collected from two students with different heights and weights, each performing five common behaviors: drinking water, picking up, clapping hands, standing up, and walking. Each student performed the five types of activities separately, with 150 sets of each activity collected under the daily environment and 30 sets collected in the dark environment, totaling 1800 sets of data. As shown in Fig. 12, the preprocessed CSI waveforms and the corresponding video frames in the daily environment demonstrate clear differences among the waveforms for each type of activities, with the video frames appearing clear and complete. Fig. 13 presents the CSI waveforms and corresponding video frames in the dark environment. Compared with the data from the daily environment, the CSI waveforms exhibit minimal variation and are largely unaffected, while the video frames appear significantly darker, making it difficult to distinguish objects and individuals.

To validate the effectiveness of the proposed method, a black rectangle with a size of  $60 \times 60$  pixels was added to occlude the leg and waist regions of the subjects in the video content to simulate the occlusion of various objects in daily environments.



Fig. 14. Lower occlusion.



Fig. 15. Middle occlusion.

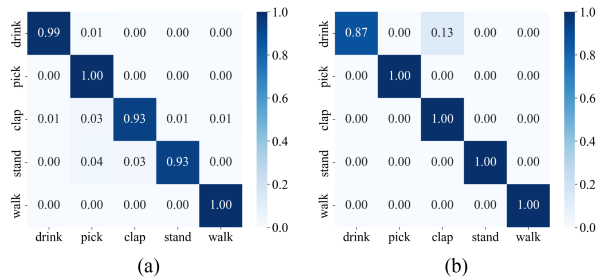


Fig. 16. Experimental results. (a) Confusion matrix of WiFi deep learning model. (b) Confusion matrix of vision deep learning model.

TABLE I  
ABLATION EXPERIMENTS OF WiFi DEEP LEARNING MODEL

| Model    | Acc (%) | Model         | Acc (%)      |
|----------|---------|---------------|--------------|
| CNN      | 95.14   | CNN+ASPP      | 96.00        |
| CNN+SE   | 95.43   | CNN+ASPP+SE   | 96.29        |
| CNN+ECA  | 95.14   | CNN+ASPP+ECA  | 95.71        |
| CNN+CBAM | 95.71   | CNN+ASPP+CBAM | <b>96.86</b> |

Lower occlusion is used to simulate the occlusion of human movements by the objects with lower heights, such as coffee tables, stools, and boxes, as shown in Fig. 14. Middle occlusion is used to simulate the occlusion of human movements by the objects with higher heights, such as dining tables, chairs, and cabinets, as shown in Fig. 15.

### B. Experimental Results

Each model was trained and tested using the daily environment dataset shown in Fig. 12, with 80% of the data used for training and 20% for testing. The WiFi deep learning model achieved an accuracy of 96.86%, and the confusion matrix is shown in Fig. 16(a). Among these activities, picking up and walking have the highest accuracy rate, reaching 100%, while clapping hands and standing up have the lowest accuracy rate of 93%. Notably, 4% of standing up were misclassified as picking up, likely due to the high similarity between standing up and the latter part of picking up.

To evaluate the contributions of each structural component in the WiFi deep learning model, we conducted ablation experiments and compared the squeeze-and-excitation (SE) and efficient channel attention (ECA) modules, as shown in Table I. The preliminary design of the CNN network has an accuracy of

TABLE II  
COMPARISON OF DIFFERENT LIGHTS

| Method     | Daily environment |              |              |              | Dark environment |              |              |              |
|------------|-------------------|--------------|--------------|--------------|------------------|--------------|--------------|--------------|
|            | Acc (%)           | Pre (%)      | Rec (%)      | F1 (%)       | Acc (%)          | Pre (%)      | Rec (%)      | F1 (%)       |
| Our WiFi   | 96.86             | 96.95        | 96.86        | 96.85        | 96.33            | 96.44        | 96.33        | 96.32        |
| Our Vision | 97.43             | 97.72        | 97.43        | 97.42        | 62.00            | 60.55        | 62.00        | 51.15        |
| WiViHAR    | <b>99.71</b>      | <b>99.72</b> | <b>99.71</b> | <b>99.71</b> | <b>99.00</b>     | <b>99.05</b> | <b>99.00</b> | <b>99.00</b> |

95.14%. When the attention mechanism was added individually, the accuracy was not significantly improved. After adding the ASPP module individually, the accuracy increased to 96.00%. Then, we further combined ASPP with different attention mechanisms, the accuracy both improved and decreased. Notable, the CNN+ASPP+CBAM model achieves the best performance, with an accuracy of 96.86%, outperforming all other variants. These results demonstrate that both ASPP and CBAM contribute positively to the model's performance and are more suitable for WiFi feature modeling in this work.

The vision deep learning model achieved an accuracy of 97.43%, and the confusion matrix is shown in Fig. 16(b). The accuracy of four activities can reach 100%, while only drinking water has an accuracy of 87%. The remaining 13% were all recognized as hand clapping, probably caused by the fact that the first half of the activity of drinking water is highly similar to the hand waving in the hand-clapping action.

To verify the effectiveness of the 3D-ASPP module in the vision deep learning model, we conducted ablation experiments. The preliminary design of the 3D-CNN network has an accuracy of 96.57%. Adding two slightly different 3D-ASPP modules increased the accuracy to 97.43%, as the final vision deep learning model. The experimental results show that 3D-ASPP module is effective for improving model performance.

After the decision fusion based on RF, our proposed method can inherit the advantages of WiFi and vision and outperform each single modality with an accuracy of 99.71%.

### C. Discussion and Analysis

1) *Comparison of Different Lights*: In order to verify the robustness of the fusion model under different lighting conditions, we collected activity data in the dark classroom. As shown in Fig. 13, the whole environment is only faintly lit by the incoming light from the window, the wall and the blackboard are vaguely visible, and the rest of the environment is darker and difficult to distinguish. The WiFi information is insensitive to the light changes and is basically unaffected. The results comparison of each modality is shown in Table II.

Specifically, in the dark environment, our vision deep learning model achieves an accuracy of only 62.00%. These results indicate that although some activities can still be accurately recognized, the recognition performance for certain activities (e.g., picking up and clapping hands) is extremely poor and unreliable, as shown in the confusion matrix in Fig. 17(a). In contrast, our WiFi deep learning model achieves 96.33% for all metrics, which is basically the same as in the daily environment, as shown in the confusion matrix in Fig. 17(b). Our WiViHAR method using the RF to fuse the WiFi and vision achieves

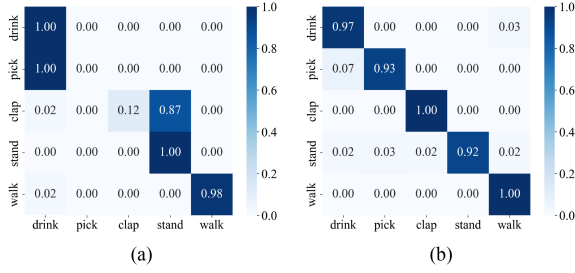


Fig. 17. Comparison of different lights. (a) Confusion matrix of the vision model under dark lighting. (b) Confusion matrix of the WiFi model under dark lighting.

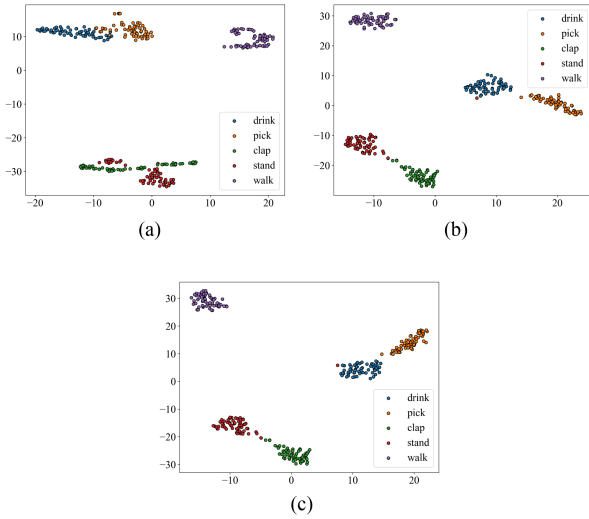


Fig. 18. T-SNE of dark environment. (a) T-SNE of the vision model under dark environment. (b) T-SNE of the WiFi model under dark environment. (c) T-SNE of the WiViHAR under dark environment.

99.00% for all metrics. This proves that our proposed fusion model has strong robustness and can achieve high recognition accuracy under different lighting conditions.

To further analyze the individual contributions of the vision and WiFi modalities in the dark environment, we performed t-distributed stochastic neighbor embedding (t-SNE) visualizations of the vision, WiFi, and fusion features, as shown in Fig. 18. As observed from Fig. 18(a), the discriminative ability of the vision features significantly degrades, with evident overlap between different classes, blurred class boundaries. In contrast, as shown in Fig. 18(b), the WiFi features maintain good separability, with clear class boundaries and minimal overlap between classes. On this basis, the fused features in Fig. 18(c) are similar to the WiFi features, while showing clearer and more compact class divisions, with increased margins between classes. The above analysis indicates that, in the dark environment, WiFi features serve as the primary support for system robustness, while vision features still provide auxiliary information for differentiating certain actions. The fusion strategy successfully integrates the complementary strengths of both modalities, validating the rationality and effectiveness of the WiViHAR design.

2) *Comparison of Fusion Methods:* To verify the performance of the RF fusion method, we chose four other common methods to compare, and the results are shown in Table III.

TABLE III  
COMPARISON OF FUSION METHODS

| Method       | Daily environment |              |              |              | Dark environment |              |              |              |
|--------------|-------------------|--------------|--------------|--------------|------------------|--------------|--------------|--------------|
|              | Acc (%)           | Pre (%)      | Rec (%)      | F1 (%)       | Acc (%)          | Pre (%)      | Rec (%)      | F1 (%)       |
| Summation    | <b>99.71</b>      | <b>99.72</b> | <b>99.71</b> | <b>99.71</b> | 96.67            | 96.78        | 96.67        | 96.66        |
| DNN          | 97.14             | 97.18        | 97.14        | 97.15        | 98.00            | 98.06        | 98.00        | 98.00        |
| Transformer  | 98.29             | 98.30        | 98.29        | 98.29        | 98.00            | 98.04        | 98.00        | 98.00        |
| SVM          | 99.14             | 99.18        | 99.14        | 99.14        | 98.00            | 98.05        | 98.00        | 98.00        |
| RF (WiViHAR) | <b>99.71</b>      | <b>99.72</b> | <b>99.71</b> | <b>99.71</b> | <b>99.00</b>     | <b>99.05</b> | <b>99.00</b> | <b>99.00</b> |

Bold values indicate the maximum value of each column.

TABLE IV  
COMPARISON OF MULTIPLE OCCLUSION

| Method       | Lower occlusion |              |              |              | Middle occlusion |              |              |              |
|--------------|-----------------|--------------|--------------|--------------|------------------|--------------|--------------|--------------|
|              | Acc (%)         | Pre (%)      | Rec (%)      | F1 (%)       | Acc (%)          | Pre (%)      | Rec (%)      | F1 (%)       |
| Summation    | 98.57           | 98.63        | 98.57        | 98.58        | <b>98.86</b>     | <b>98.92</b> | <b>98.86</b> | <b>98.87</b> |
| DNN          | 97.43           | 97.44        | 97.43        | 97.43        | 97.14            | 97.18        | 97.14        | 97.15        |
| Transformer  | 98.00           | 98.01        | 98.00        | 98.00        | 98.00            | 98.03        | 98.00        | 97.99        |
| SVM          | 96.00           | 96.52        | 96.00        | 96.02        | 96.00            | 96.52        | 96.00        | 96.02        |
| RF (WiViHAR) | <b>99.14</b>    | <b>99.18</b> | <b>99.14</b> | <b>99.15</b> | <b>98.86</b>     | <b>98.92</b> | <b>98.86</b> | 98.86        |

Bold values indicate the maximum value of each column.

The summation method is a simple addition of the outputs from two network models to obtain the final result. It performs very well in daily environments, which is on par with the RF fusion method, because both the WiFi model and the vision model have high accuracy at this time. However, in the dark environment, the accuracy of the vision model decreases significantly, which negatively impacts the summation result.

The deep neural network (DNN) method performs poorly in the daily environment, even lower than that of the single vision model. This may be because DNN fails to effectively capture the complementary information between the two submodels. Nevertheless, all metrics improve to 98.00% in the dark environment, showing that DNN can compensate for partial information loss through nonlinear mappings.

The Transformer method achieves 98.29% for all metrics in the daily environment and 98.00% for all metrics in the dark environment, showing stable performance. Thanks to its feature modeling capability and self-attention mechanism, Transformer can capture deep dependencies between submodels. However, due to its complex structure and high data demand, its performance is lower than the RF fusion method.

The support vector machine (SVM) method performs well in the daily environment, achieving 99.14% for all metrics, which is close to that of the RF fusion method. This is mainly due to the excellent boundary delineation ability of SVM in high-dimensional space. However, all metrics drop to 98.00% in the dark environment, indicating a limited generalization ability when handling complex or ambiguous features.

Overall, the RF fusion method achieves the best performance in both environments. This can be attributed to the RF's strong feature modeling ability, which enables it to better balance the information from multiple submodels. It can be concluded that our proposed fusion model outperforms the four common methods in terms of overall performance.

3) *Comparison of Multiple Occlusion:* In order to verify the robustness of the proposed fusion method in various occlusion environments, we simulate the different occlusion by adding black rectangular blocks, as shown in Figs. 14 and 15.

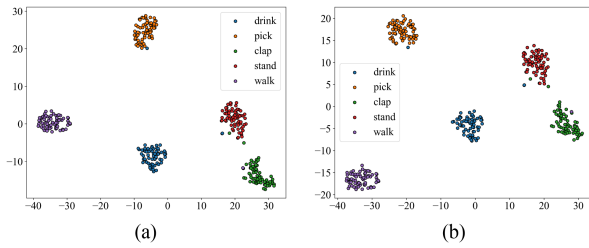


Fig. 19. T-SNE of multiple occlusion. (a) T-SNE of the WiViHAR under lower occlusion. (b) T-SNE of the WiViHAR under middle occlusion.

TABLE V  
COMPARISON OF PUBLIC METHODS

| Method     | Modal       | Acc (%)      | Pre (%)      | Rec (%)      | F1 (%)       |
|------------|-------------|--------------|--------------|--------------|--------------|
| LeNet-5    | WiFi        | 93.00        | 93.22        | 93.00        | 92.96        |
| ResNet18   | WiFi        | 94.33        | 94.47        | 94.33        | 94.35        |
| Our WiFi   | WiFi        | 96.33        | 96.44        | 96.33        | 96.32        |
| C3D        | Vision      | 54.00        | 46.06        | 54.00        | 45.77        |
| Our Vision | Vision      | 62.00        | 60.55        | 62.00        | 51.15        |
| WiVi       | WiFi+Vision | 97.33        | 97.37        | 97.33        | 97.33        |
| WiViHAR    | WiFi+Vision | <b>99.00</b> | <b>99.05</b> | <b>99.00</b> | <b>99.00</b> |

As can be seen from Table IV, in the case of lower occlusion, the methods of summation and SVM have a significant decrease compared to the normal case, and the methods of DNN, Transformer, and RF have only small fluctuations. The RF fusion method still obtains the best result, achieving 99.14% for all metrics. In the case of middle occlusion, the SVM method has a significant decrease compared to the normal case, and the DNN, Transformer, and RF methods have only small fluctuations. The summation method has a partial increase and achieves the same best results as the RF fusion method, reaching 98.86% for all metrics.

To further improve practical relevance, we also conducted a physical occlusion experiment using common boxes and chairs. In the case of box occlusion, the RF fusion method has an accuracy of 99.43%. In the case of chair occlusion, the RF fusion method has an accuracy of 98.57%. Although the results fluctuated slightly, they were still better than the other four methods. It can be seen that our proposed fusion model has strong robustness and can achieve excellent performance in multiple occlusion environments.

In addition, we performed t-SNE visualizations of the fused features under both lower and middle occlusion conditions, as shown in Fig. 19. It can be observed that the clustering performance under these occlusion scenarios is superior to that in the dark environment. The class boundaries are more clearly delineated, the clustering structures are more compact, and the cluster shapes tend to be ideally circular, indicating improved separability. These results suggest that when vision perception is affected by occlusion, the fusion strategy remains effective and can maintain or even enhance the overall discriminative capability. This further verifies the effectiveness and robustness of WiViHAR in different types of perception defect scenarios.

4) *Comparison of Public Methods:* In order to verify the performance of WiViHAR, we chose several publicly available methods for comparison in the dark environment, and the results are shown in Table V.

TABLE VI  
EFFICIENCY EVALUATION

| Module | Params (M)  | FLOPs (M)   | Inference time (ms) |
|--------|-------------|-------------|---------------------|
| WiFi   | 1.52        | 808.40      | 10.26               |
| Vision | 29.63       | 4874.41     | 115.02              |
| Fusion | 484 (nodes) | 227 (comps) | 0.04                |

LeNet-5 [33] and ResNet18 [34] both are designed for image recognition. LeNet-5 achieved an accuracy of 93.00%, and ResNet18 achieved an accuracy of 94.33%. Their results are lower than those achieved by our proposed WiFi deep learning model. C3D [14] is a network specifically designed for video recognition, achieved an accuracy of only 54.00% in the dark environment, which is also lower than the vision deep learning model presented in this work. WiVi [35] is a multimodal recognition approach proposed by Zou et al. [35], achieved an accuracy of 97.33%, showing an improvement over single-modality methods but still slightly lower than WiViHAR.

These comparisons with various publicly available methods demonstrate that our proposed fusion model offers superior overall performance, achieving higher accuracy and outperforming the other methods.

5) *Efficiency Evaluation:* To evaluate the WiViHAR's efficiency, we provide a detailed analysis of each module, as shown in Table VI. The WiFi sensing module maintains a lightweight design with only 1.52 M parameters and 808.40 M floating point operations (FLOPs), achieving a fast inference time of 10.26 ms. In contrast, the vision sensing module requires 29.63 M parameters and 4874.41 M FLOPs, and due to the extraction of high-dimensional spatiotemporal features, the inference time is significantly longer at 115.02 ms. The multimodal fusion module is implemented via the RF with 484 nodes and 227 comparisons, incurring negligible computational overhead (0.04 ms). These evaluations were conducted on a laptop equipped with an Intel Core i5-7200 U CPU and an NVIDIA GeForce 940MX GPU. Overall, the WiViHAR achieves a balance between recognition accuracy and computational efficiency, making it feasible for real-time deployment in practical scenarios.

## V. CONCLUSION

In this article, we proposed a WiFi and vision multimodal fusion method for HAR, WiViHAR, based on improved deep learning. First, WiFi information and vision information are simultaneously captured using a COTS WiFi device and a high-definition camera. Second, dedicated deep learning network models are designed for the WiFi sensing module and the vision sensing module, respectively, to extract more effective features. Among them, we design the ASPP and CBAM in the WiFi deep learning model and extend the ASPP to 3-D structures in the vision deep learning model. Finally, we utilize the RF to fuse the outputs of the two models to obtain the final recognition result. Extensive experiments conducted in real-world environments demonstrated that our proposed method obtained an accuracy of 98.57% or higher under various conditions, which is significantly better than the single-modal methods for WiFi or vision and has strong robustness.

Despite promising results, the small size of the current dataset inevitably raises concerns about overfitting and generalization ability. Therefore, future work will focus on expanding the dataset, recruiting more participants with different body shapes, ages, and activity patterns, and collecting data in a wider range of environments, thereby facilitating the deployment of the proposed method in real-world scenarios. At the same time, improving model interpretability is also an important direction. We attempted to visualize the feature maps generated by ASPP and CBAM, but without achieving meaningful or effective results. One possible reason is that the features learned by these modules are themselves very complex and abstract, resulting them difficult to project into a visually interpretable 2-D space. In the future, we plan to explore more advanced interpretability techniques tailored for multimodal data, thereby enhancing the transparency of the proposed method.

Furthermore, future work will also consider fusion with other modalities (e.g., inertial measurement units or audio signals). For example, inertial measurement units embedded in smartphones and wearable devices can provide fine-grained motion data to complement WiFi signals, while audio signals may capture activity-related acoustic cues in scenarios where vision information is insufficient. However, integrating these modalities faces numerous challenges, including temporal synchronization, feature alignment, and privacy issues. Under these practical limitations, exploring more fusion methods (e.g., feature fusion or hybrid fusion) may provide more robust and versatile HAR systems.

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